

IGE-LIO: Intensity Gradient Enhanced Tightly Coupled LiDAR–Inertial Odometry

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Abstract—Simultaneous localization and mapping (SLAM) plays an important role in the state estimation of mobile robots. Most popular LiDAR SLAM (L-SLAM) methods extract feature points only from the geometric structure of the environment, which can result in inaccurate localization in degenerated scenarios. In this article, we present a novel framework for LiDAR intensity gradient enhanced tightly coupled LiDAR–inertial odometry (IGE-LIO). The framework proposes a novel LiDAR intensity gradient-based feature extraction approach for accurate pose estimation, overcoming the challenges faced by L-SLAM in degenerated environments. After computing the intensity gradient of each LiDAR point, we dynamically extract intensity edge points (IEPs) from texture information. In addition, we extract geometric planar points (GPPs) and geometric edge points (GEPs) based on geometric information. Then, the error analysis is performed on each type of feature points, and the weighting functions are designed to correct measurement noise and mitigate biases introduced by the additional uncertainty in feature extraction. Subsequently, an iterative extended Kalman filter (IEKF) framework is constructed by combining residuals from point-to-plane and point-to-edge associations. Finally, extensive experiments are conducted in indoor, outdoor, and LiDAR degenerated scenarios. The results demonstrate the significantly improved robustness and accuracy of our proposed method compared with the existing geometric-only methods, especially in LiDAR degenerated scenarios.

Index Terms—Degenerated environments, intensity gradient, localization, simultaneous localization and mapping (SLAM), weighting function.

I. INTRODUCTION

SIMULTANEOUS localization and mapping (SLAM) is a fundamental capability of autonomous mobile robots, such as unmanned ground vehicles (UGVs) and unmanned

aerial vehicles (UAVs). It allows simultaneous perception of the environment and estimation of the pose, enabling incremental mapping in unknown environments [1], [2]. This significantly reduces the dependence on satellite positioning and previous maps for autonomous navigation. SLAM can be further divided into visual SLAM (V-SLAM) and LiDAR SLAM (L-SLAM) based on the perception system used. Visual odometry (VO) methods use RGB textures [3], [4], [5] and offer the advantages of low cost and lightweight. However, they are susceptible to changes in lighting conditions. In contrast, L-SLAM is more accurate and robust, regardless of environmental changes such as weather and illumination.

Traditional L-SLAM [6], [7], [8], [9], [10], [11] focuses only on the geometric information in the environment, which can achieve ideal results in environments with rich geometric structures. However, degeneracy poses a significant challenge in L-SLAM that cannot be disregarded. For instance, in elongated corridors or environments dominated by single-sided walls, robots struggle to accurately estimate their motion along the direction of the wall. To address this issue, new types of perception sensors, such as cameras, are often integrated into the SLAM framework, but this requires additional cost. In contrast, our proposed method uses the intensity information from LiDAR to extract additional constraints from texture information of the environment, which mitigates the problem of geometric degradation.

Note that the information from the intensity channel is related to the reflectivity of material, measured distance, and incident angle, which can be useful for localization and object recognition. In contrast to the passive exposure of cameras, the intensity images actively measured by the LiDAR are nearly independent of illumination, offering great potential for mapping object textures. Furthermore, the intensity values exhibit significant gradient invariance at boundaries, facilitating the extraction of edge features. By appropriately combining the geometric features with the texture features derived from the LiDAR intensity, even environments lacking geometric features can be effectively handled [12]. Hewitt and Marshall [13] analyzed the physical model of intensity measurement and validated through simulation that fusing intensity measurement values can achieve accuracy comparable to SLAM even in situations with fewer feature points. Therefore, we argue that a robust L-SLAM framework should take the intensity information into account. Current L-SLAM systems that incorporate intensity information require intensity calibration and reflectivity modification [14], [15], [16]. However, calculating the accurate incident angle is exceedingly

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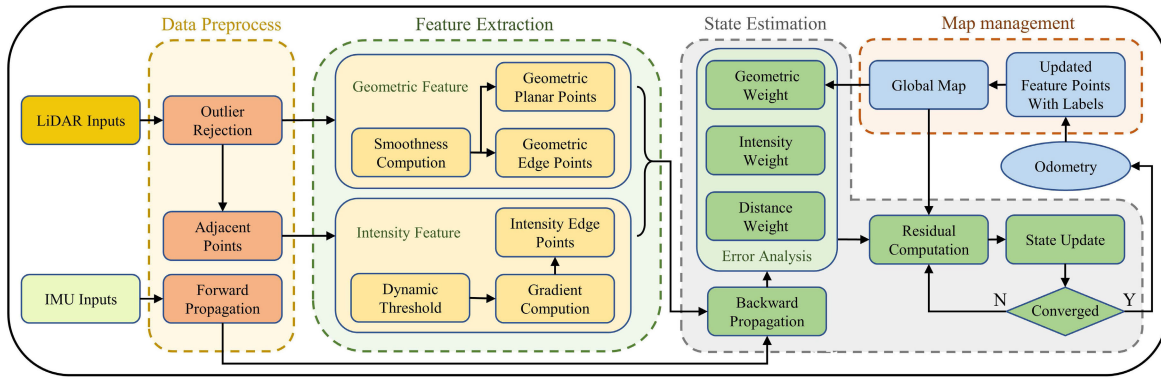


Fig. 1. System overview of IGE-LIO.

difficult. In contrast, our proposed method uses intensity gradients to extract additional feature points without intensity correction and uses designed weighting functions to improve robustness and accuracy.

In this article, we propose a framework for pose estimation that uses an iterative extended Kalman filter (IEKF) to tightly couple the inertial measurement unit (IMU), geometric residual, and texture residual. The framework of the system is illustrated in Fig. 1. Geometric planar points (GPPs), geometric edge points (GEPs), and intensity edge points (IEPs) are extracted based on the geometric and intensity information. In addition, the error analysis is performed for each type of feature point using a designed weighting function. Residuals are then computed separately for different categories of feature points, and the state is updated iteratively until convergence is achieved. Finally, feature points with their labels are stored in the global map. The results demonstrate that the proposed method not only achieves accurate, robust, and real-time localization and mapping in LiDAR degenerated scenarios but also surpasses traditional L-SLAM, reaching results comparable to SLAM methods that incorporate visual information. The main contributions of this article are listed as follows.

- 1) A robust LIO framework for tightly coupling LiDAR geometric and texture information with an IMU via the IEKF for pose estimation is presented. The framework can achieve optimal results for rotating LiDARs and solid-state LiDARs.
- 2) A novel feature extraction method is proposed, which not only extracts GPPs and GEPs but also computes the intensity gradient of each LiDAR point without calibrating the intensity value and dynamically selects IEPs using a dynamic threshold.
- 3) To distinguish and weigh the quality of each feature point for pose optimization, we introduce a new weighting function that jointly fuses geometric and intensity information for GPPs, GEPs, and IEPs.

II. RELATED WORK

A. Geometric-Only L-SLAM

Due to the large amount of raw LiDAR data, processing these point clouds directly in conventional L-SLAM systems is time-consuming. To ensure real-time performance of

L-SLAM, most existing L-SLAM approaches extract feature points from the geometric structures in the environment. Zhang and Singh [6] proposed LOAM, which extracts edge and planar feature points based on the local smoothness of each point. To accelerate computation speed, Shan and Englot [7] proposed LeGO-LOAM, which performs point cloud segmentation and clustering to extract ground feature points and applies a two-step optimization for pose estimation. LOAM-Livox proposed by Lin and Zhang [8] addresses the issues of limited field of view (FoV) and the irregular scanning model in solid-state LiDAR.

Considering the short-term reliability of the IMU, many SLAM frameworks actively incorporate IMU to improve accuracy and robustness. Shan et al. [9] proposed LIO-SAM, which uses ISAM2 to construct a factor graph, which facilitates the integration of various sensors into the system. Xu and Zhang [10] proposed Fast-LIO, which represents a new formula to compute the Kalman gain. Furthermore, Xu et al. [11] proposed Fast-LIO2, which uses an ikd-tree data structure to represent the large and dense point clouds. Although these methods are applied well in most scenarios, the degenerated environments still pose significant challenges.

B. Intensity-Fusion L-SLAM

Recently, several studies have attempted to improve accuracy by leveraging LiDAR intensity information. Lu et al. [17] proposed DeepICP, an end-to-end learning approach for 3-D point cloud registration. Chen et al. [18] proposed an end-to-end CNN framework for loop closure detection, which uses both intensity and geometric information from LiDAR data as inputs to generate features. Similarly, Bârsan et al. [19] capitalized on LiDAR intensity information as input. Although learning-based methods can avoid additional processing of intensity information, it is still imperative to collect, label, and train the intensity information in advance in practice, which requires a significant amount of resources. Furthermore, the performance under environment changes may not remain consistent.

There are some studies that use LiDAR intensity for pose estimation without deep learning. The effectiveness of using intensity values for position recognition has also been demonstrated [20], [21]. After Kim and Kim [22] developed SC, Wang et al. [23] proposed ISC, which encodes

the corresponding intensity information into a 2-D matrix. Di Giammarino et al. [24] showed that visual localization techniques applied to LiDAR intensity can be considered as a feasible loop-closure detection method. In [25], visual features are extracted from LiDAR intensity images and tracked frame by frame to achieve VO. In addition, ORB feature descriptors are also extracted from intensity images [26], [27]. Wang et al. [14] proposed Intensity-SLAM, which uses geometric and calibrated intensity information to build a reflectivity distribution map. The method minimizes the reflectivity residual constructed based on trilinear interpolation and the geometric residual of pose estimation. Zhang et al. [15] proposed RI-LIO, which generates the reflectivity images without empty holes via CPBRA to construct the photometric residual. Li et al. [16] extracted feature points based on geometric and intensity information.

III. METHODOLOGY

A. Kinematic Model

1) *State Transition Model*: Assuming that IMU is rigidly attached to LiDAR with an initial extrinsic ${}^I\mathbf{T}_L = ({}^I\mathbf{R}_L, {}^I\mathbf{p}_L)$, where L represents the LiDAR coordinate system and I represents the IMU coordinate system, which also serves as the body coordinate system. We consider the first frame of the IMU coordinate system as the global coordinate system, denoted as G . ${}^G\mathbf{T}_I = ({}^G\mathbf{R}_I, {}^G\mathbf{p}_I)$ represents the current position and orientation of the IMU in the global frame. The discrete model can be represented

$$\mathbf{x}_{i+1} = \mathbf{x}_i \boxplus (\Delta t \mathbf{f}(\mathbf{x}_i, \mathbf{u}_i, \boldsymbol{\omega}_i)) \quad (1)$$

where i is the index of IMU data, $\mathbf{u}_i$ and $\boldsymbol{\omega}_i$ represent the input and noise, respectively. The symbol $\boxplus$ denotes the generalized operator defined in [10], as follows:

$$\begin{aligned} \boxplus : \mathcal{M} \times \mathbb{R}^n &\rightarrow \mathcal{M} \\ \mathcal{M} = SO(3) : \mathbf{R} \boxplus \mathbf{r} &= \mathbf{R} \text{Exp}(\mathbf{r}) \\ \mathcal{M} = \mathbb{R}^n : \mathbf{a} \boxplus \mathbf{b} &= \mathbf{a} + \mathbf{b} \end{aligned} \quad (2)$$

where $\mathcal{M}$ is the n -dimensional manifold, $\mathbb{R}^n$ is its tangent space, and $\text{Exp}(\mathbf{r})$ denotes the exponential map. It establishes a bijective mapping from a local neighborhood on $\mathcal{M}$ to $\mathbb{R}^n$ through $\boxplus$. Thus, for a compound manifold $\mathcal{M} = SO(3) \times \mathbb{R}^n$

$$\begin{bmatrix} \mathbf{R} \\ \mathbf{a} \end{bmatrix} \boxplus \begin{bmatrix} \mathbf{r} \\ \mathbf{b} \end{bmatrix} = \begin{bmatrix} \mathbf{R} \boxplus \mathbf{r} \\ \mathbf{a} + \mathbf{b} \end{bmatrix}. \quad (3)$$

2) *Measurement Model*: The measurement noise ${}^L\mathbf{n}_j$ of the j th point ${}^L\mathbf{p}_j$ in the k th scan is influenced by both LiDAR ranging noise and beam-directing noise. The measurement contributed by the point ${}^L\mathbf{p}_j$ can be represented

$$\begin{aligned} 0 &= \mathbf{h}_j(\mathbf{x}_k, {}^L\mathbf{n}_j) \\ &= \mathbf{G}_j({}^G\mathbf{T}_k {}^I\mathbf{T}_L ({}^L\mathbf{p}_j + {}^L\mathbf{n}_j) - {}^G\mathbf{q}_j). \end{aligned} \quad (4)$$

For planar points, $\mathbf{G}_j = \mathbf{u}_j^T$, where $\mathbf{u}_j$ is the normal vector. For edge points, $\mathbf{G}_j = [\mathbf{u}_j]_{\wedge}$, where $\mathbf{u}_j$ represents the edge orientation. ${}^G\mathbf{q}_j$ represents the points on the corresponding plane or edge.

B. Data Preprocess

1) *Forward Propagation*: The robotic pose at each time step is computed through numerical integration. After processing the last LiDAR scan, the updated state is $\bar{\mathbf{x}}_{k-1}$ with the covariance of $\bar{\mathbf{P}}_{k-1}$ being obtained. Forward propagation is performed upon the arrival of an IMU measurement

$$\begin{aligned} \hat{\mathbf{x}}_{i+1} &= \hat{\mathbf{x}}_i \boxplus (\Delta t \mathbf{f}(\hat{\mathbf{x}}_i, \mathbf{u}_i, \mathbf{0})); \hat{\mathbf{x}}_0 = \bar{\mathbf{x}}_{k-1} \\ \hat{\mathbf{P}}_{i+1} &= \mathbf{F}_{\hat{\mathbf{x}}_i} \hat{\mathbf{P}}_i \mathbf{F}_{\hat{\mathbf{x}}_i}^T + \mathbf{F}_{\mathbf{w}_i} \mathbf{Q}_i \mathbf{F}_{\mathbf{w}_i}^T; \hat{\mathbf{P}}_0 = \bar{\mathbf{P}}_{k-1} \end{aligned} \quad (5)$$

where $\mathbf{Q}_i$ is the covariance of the process noise $\mathbf{w}_i$, and $\hat{\mathbf{x}}_k$ and $\hat{\mathbf{P}}_k$ represent the propagated state. $\tilde{\mathbf{x}}_k = \mathbf{x}_k \boxminus \hat{\mathbf{x}}_k$ represents the error between the true value $\mathbf{x}_k$ and the final estimated value $\hat{\mathbf{x}}_k$. The matrices $\mathbf{F}_{\hat{\mathbf{x}}_i}$ and $\mathbf{F}_{\mathbf{w}_i}$ are

$$\begin{aligned} \mathbf{F}_{\hat{\mathbf{x}}_i} &= \left. \frac{\partial (\mathbf{x}_{i+1} \boxminus \hat{\mathbf{x}}_{i+1})}{\partial \tilde{\mathbf{x}}_i} \right|_{\tilde{\mathbf{x}}_i=\mathbf{0}, \mathbf{w}_i=\mathbf{0}} \\ \mathbf{F}_{\mathbf{w}_i} &= \left. \frac{\partial (\mathbf{x}_{i+1} \boxminus \hat{\mathbf{x}}_{i+1})}{\partial \mathbf{w}_i} \right|_{\tilde{\mathbf{x}}_i=\mathbf{0}, \mathbf{w}_i=\mathbf{0}}. \end{aligned} \quad (6)$$

2) *Outlier Rejection*: Before processing the point cloud, outlier rejection is a necessary step to improve localization accuracy and system robustness. First, LiDAR points may be occluded by the equipment due to the influence of equipment size and LiDAR installation position, which lead to the ghosting in the generated point cloud map. In addition, points too close to the LiDAR cause measurement distortion. Simultaneously, points too far from the LiDAR are often sparse and may have significant noise. Therefore, the confidence in points that are too close or far diminishes. Taking into account the size of the equipment and the type of LiDAR, we exclude points that are beyond the range 0.5–150 m from the LiDAR.

Moreover, LiDAR points with extremely large or small intensity values are unable to effectively represent texture information. Low intensity values may result in a disproportion between signal and noise, thereby inaccurately representing texture information. On the other hand, excessively high intensity values cannot accurately reflect subtle texture information on the object surface. To reduce the impact of these points, we set the threshold of outlier rejection for intensity values at $\eta < 15$ or $\eta > 240$, given that pixel values in images typically range from 0 to 255. Although these points are excluded from the extraction of intensity feature points, they are still used for the extraction of geometric feature points. Outlier rejection also prevents unnecessary computations on these points in subsequent steps to conserve computational resources.

3) *Adjacent Points Selection*: The proposed method directly uses the intensity information without requiring intensity calibration or reflectivity modification. To ensure that the calculated intensity gradient of the LiDAR points is more accurate, it is essential to first select the adjacent points of the LiDAR point.

The scan lines of the rotating LiDARs are typically horizontal and parallel to each other. Relying on scan lines alone is unable to extract adjacent point sets in other directions, resulting in the final IEPs being extracted only from edges in a single direction. We use intensity images to select adjacent points in other directions. As shown in Fig. 2, the LiDAR

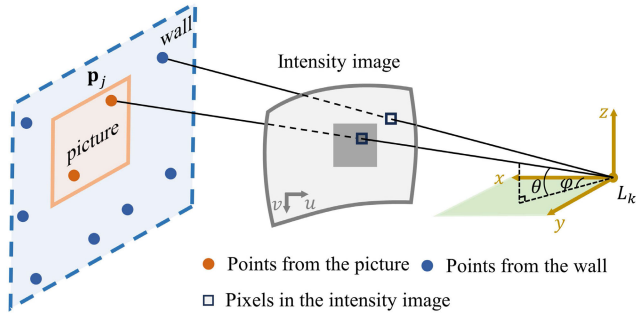


Fig. 2. Process of generating the intensity image. The points with different colors represent varying intensity values.

point $\mathbf{p}_j$ is transformed to image coordinates (u, v)

$$\begin{aligned} \varphi &= \arctan(y, x); \quad \theta = \arctan\left(z, \sqrt{x^2 + y^2}\right) \\ u &= \frac{W}{\varphi_{\text{fov}}}(\varphi^{\text{right}} - \varphi); \quad v = \frac{H}{\theta_{\text{fov}}}(\theta^{\text{up}} - \theta) \end{aligned} \quad (7)$$

where (R, θ, φ) represents spherical coordinates. W and H denote the width and height of the image, respectively. $\varphi^{\text{fov}} = \varphi^{\text{right}} - \varphi^{\text{left}}$ represents the horizontal FoV of the LiDAR. $\theta^{\text{fov}} = \theta^{\text{up}} - \theta^{\text{down}}$ represents the vertical FoV of the LiDAR. For the rotating LiDARs, the scan line s_j of the point $\mathbf{p}_j$ can represent the vertical coordinate, i.e., $v = s_j$. In addition, the horizontal FoV is generally 360° , denoted as $\varphi^{\text{fov}} = 2\pi$, $\varphi^{\text{right}} = \pi$. The pixel value is determined by the intensity value η_j .

The LiDAR points located on the boundary between two objects with different intensity exhibit a large difference on one side and a small difference on the other side. If there are enough valid points within a certain distance of the point $\mathbf{p}_j$, we select and denote two sets of adjacent points: $\overline{\mathcal{P}}_{\mathcal{M}}$ with M points on one side of the point $\mathbf{p}_j$, and $\overline{\mathcal{P}}_{\mathcal{N}}$ with N points on the opposite side. In addition, $\overline{\mathcal{P}}_{\mathcal{M}}$ represents the set of adjacent points with a significant intensity difference from point $\mathbf{p}_j$, while $\overline{\mathcal{P}}_{\mathcal{N}}$ represents the set of adjacent points with intensity values close to that of point $\mathbf{p}_j$. The LiDAR points are represented by the corresponding nonhole pixels when using intensity images. Due to the low vertical resolution of the rotating LiDARs, it is sufficient to compare with only adjacent nonhole pixels in the previous row, i.e., $M = 1$ and $N = 0$, as shown in Fig. 3(a).

The scan lines of the mainstream solid-state LiDARs with the irregular scanning model intersect with each other, enabling the extraction of the adjacent point sets from edges in multiple direction. Thus, each point only needs to determine its adjacent point sets $\overline{\mathcal{P}}_{\mathcal{M}}$ and $\overline{\mathcal{P}}_{\mathcal{N}}$ along the scan line, as shown in Fig. 3(c).

C. Feature Extraction

1) *GPPs and GEPs*: Using a similar approach to LOAM, geometric feature points are extracted, and the smoothness of each LiDAR point is calculated as follows:

$$c_j = \frac{1}{|N| \cdot \|\mathbf{p}_j\|} \left\| \sum_{\mathbf{p}_i \in \mathcal{P}_j, i \neq j} (\mathbf{p}_i - \mathbf{p}_j) \right\| \quad (8)$$

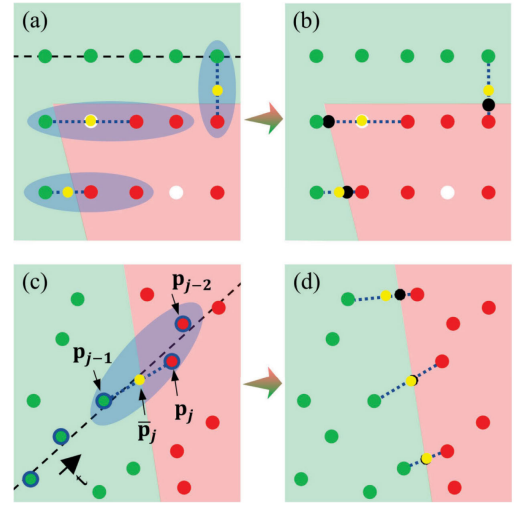


Fig. 3. Illustration of IEPs' extraction. (a) and (b) Applicable to the rotating LiDARs. (c) and (d) Applicable to the solid-state LiDARs. The green and red points have different intensity values. The white points represent empty holes. The yellow and black points represent the extracted and virtual IEPs, respectively.

where $\mathbf{p}_j$ represents the j th point in the LiDAR frame (L frame), $\mathcal{P}_j$ is the set of points surrounding $\mathbf{p}_j$, and N is the number of points in the $\mathcal{P}_j$ set.

The classification of points is based on their smoothness, with points that have higher smoothness being classified as edge feature points, and those with lower smoothness being classified as planar feature points. In the end, we obtain two sets of geometric points for the current scan: $\mathcal{P}_{\text{GPP}}$ represents the geometric planar point set, and $\mathcal{P}_{\text{GEP}}$ represents the geometric edge point set.

2) *Intensity Edge Points*: The proposed method extracts IEPs from boundaries based on intensity gradients. If the intensity value of the point is large, the intensity values of the surrounding points exhibit larger gradient fluctuations, which makes it more likely for that point to be mistakenly identified as an IEP. The proposed method adopts dynamic thresholds to reduce the occurrence of this situation. Given the current intensity value of the LiDAR point η_j , we can obtain

$$\begin{aligned} \lambda_j^{\mathcal{M}} &= \mathbf{f}_{\mathcal{M}}(\eta_j) = \min(\mathbf{C}_{\mathcal{M}}, \mathbf{V}_{\mathcal{M}} \times \eta_j) \\ \lambda_j^{\mathcal{N}} &= \mathbf{f}_{\mathcal{N}}(\eta_j) = \max(\mathbf{C}_{\mathcal{N}}, \mathbf{V}_{\mathcal{N}} \times \eta_j) \end{aligned} \quad (9)$$

where $\lambda_j^{\mathcal{M}}$ and $\lambda_j^{\mathcal{N}}$ are the thresholds used to determine whether the points in $\overline{\mathcal{P}}_{\mathcal{M}}$ and $\overline{\mathcal{P}}_{\mathcal{N}}$, respectively, are valid adjacent points of point $\mathbf{p}_j$. $\mathbf{C}_{\mathcal{M}}$ and $\mathbf{C}_{\mathcal{N}}$ are constants, and $\mathbf{V}_{\mathcal{M}}$ and $\mathbf{V}_{\mathcal{N}}$ are the corresponding line slopes; see Fig. 4.

If the absolute value of intensity difference between the each point in $\overline{\mathcal{P}}_{\mathcal{N}}$ and $\mathbf{p}_j$ is smaller than $\lambda_j^{\mathcal{N}}$, and the absolute value of intensity difference between each point in $\overline{\mathcal{P}}_{\mathcal{M}}$ and $\mathbf{p}_j$ is greater than $\lambda_j^{\mathcal{M}}$, then the intensity gradient of point $\mathbf{p}_j$ can be calculated as

$$G_j = \left| \frac{1}{N+1} \left(\eta_j + \sum_{\check{\mathbf{p}}_m^{\mathcal{M}} \in \overline{\mathcal{P}}_{\mathcal{M}}} \eta_m \right) - \frac{1}{M} \sum_{\check{\mathbf{p}}_n^{\mathcal{N}} \in \overline{\mathcal{P}}_{\mathcal{N}}} \eta_n \right| \quad (10)$$

where $\check{\mathbf{p}}_m^{\mathcal{M}}$ and $\check{\mathbf{p}}_n^{\mathcal{N}}$ represent the m th and n th points in $\check{\mathcal{P}}_{\mathcal{M}}$ and $\check{\mathcal{P}}_{\mathcal{N}}$, respectively. The point $\mathbf{p}_j$ with $G_j > G_{\text{max}}$ is

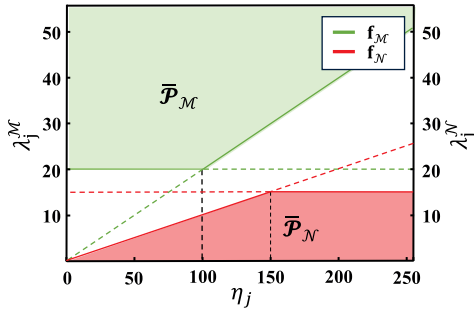


Fig. 4. Illustration of dynamic threshold.

considered as candidate for IEP, where $G_{\max}$ is the set intensity gradient threshold.

The final extracted IEP is determined using the following equation:

$$\bar{\mathbf{p}}_j = \frac{1}{2}(\check{\mathbf{p}}_0^{\mathcal{M}} + \mathbf{p}_j) \quad (11)$$

where $\check{\mathbf{p}}_0^{\mathcal{M}}$ represents the point in $\check{\mathcal{P}}_{\mathcal{M}}$ closest to $\mathbf{p}_j$. As shown in Fig. 3(b) and (d), the extracted yellow feature points deviate from the virtual black IEPs.

After processing all the points, the set of final extracted IEPs for the current frame is obtained and denoted as $\mathcal{P}_{\text{IEP}}$. Subsequently, these three types of feature points are combined to form a new scan of feature points, denoted as $\check{\mathcal{P}} = \{\mathcal{P}_{\text{GPP}}, \mathcal{P}_{\text{GEP}}, \mathcal{P}_{\text{IEP}}\}$.

D. State Estimation

1) *Backward Propagation*: Through backward propagation, we can obtain the relative pose of the IMU between t_j and t_k , denoting it as ${}^{L_k}\check{\mathbf{T}}_{I_j} = ({}^{L_k}\check{\mathbf{R}}_{I_j}, {}^{L_k}\check{\mathbf{p}}_{I_j})$. In this case, the j th point ${}^{L_j}\check{\mathbf{p}}_j$ is projected to the end of the scan

$${}^{L_k}\check{\mathbf{p}}_j = {}^L\mathbf{T}_L^{k-1} {}^{L_k}\check{\mathbf{T}}_{I_j} {}^L\mathbf{T}_L^{k-1} {}^{L_j}\check{\mathbf{p}}_j \quad (12)$$

where ${}^{L_j}\check{\mathbf{p}}_j \in \check{\mathcal{P}}$, and ${}^L\mathbf{T}_L^{k-1}$ represents the extrinsic obtained after the previous optimization. In this article, ${}^{L_k}\check{\mathbf{p}}_j$ is denoted as ${}^{L_j}\check{\mathbf{p}}_j$ for simplicity.

2) *Error Analysis*: The proposed method designs individual weighting functions for feature points based on their geometric and intensity information and uses weight values to quantify the quality of points and gives greater importance to higher quality feature points in pose optimization. Furthermore, we design the weighting functions by referring to the sigmoid function. The weight value w_P increases with the increase in the positive factor x_P , while the weight value w_N decreases with the increase in the negative factor x_N

$$w_P = \frac{\lambda_1}{1.0 + e^{-\lambda_2 x_P}}, \quad w_P \in \left(\frac{1}{2}\lambda_1, \lambda_1\right) \\ w_N = \frac{\lambda_1}{2.0 - e^{-\lambda_2 x_N}}, \quad w_N \in \left(\frac{1}{2}\lambda_1, \lambda_1\right) \quad (13)$$

where λ_1 and λ_2 are hyperparameters; λ_1 is set to 2 to constrain the weight within the range 1–2, and λ_2 can control the change rate of the factor, typically set to 1.

According to the definition of smoothness, a smaller smoothness value c_j of the point ${}^{L_j}\check{\mathbf{p}}_j$ indicates a higher likelihood that it originates from a plane. Considering the intensity information, let $\mathcal{P}_j$ represent the set of points consisting of the GPP and its surrounding planar points. Planar points with almost the same intensity are more likely to belong to the same plane compared with points with irregular intensity distribution. Therefore, the intensity distribution of points in $\mathcal{P}_j$ is calculated using the same method as in [16], which can provide weighted information for evaluating the quality of the planar points

$$\bar{\eta} = \frac{1}{N} \sum_{{}^{L_j}\check{\mathbf{p}}_i \in \mathcal{P}_j} \eta_i; \sigma_{\eta}^2 = \frac{1}{N-1} \sum_{{}^{L_j}\check{\mathbf{p}}_i \in \mathcal{P}_j} (\eta_i - \bar{\eta})^2 \quad (14)$$

where η_i represents the intensity value of the point ${}^{L_j}\check{\mathbf{p}}_i \in \mathcal{P}_j$, $\bar{\eta}$ denotes the mean intensity of the adjacent point set, and σ_{η}^2 represents the variance of the intensity distribution.

Due to the greater impact of smoothness on GPPs, the smoothness-related parameter λ_2 is set to 2. Therefore, with the combined influence of smoothness and intensity distribution, the weight of GPP w_j^{GPP} can reflect the quality of the planar point

$$w_j^{\text{GPP}} = \frac{2.0}{2.0 - e^{-2\sigma_{\eta}}} \times \frac{2.0}{2.0 - e^{-c_j}}. \quad (15)$$

The extraction of GEPs is based on evaluating the differences between point ${}^{L_j}\check{\mathbf{p}}_j$ and the surrounding points using geometric information to extract points with high smoothness. The formula for the weight of GEP w_j^{GEP} is

$$w_j^{\text{GEP}} = \frac{2.0}{2.0 - e^{-c_j}}. \quad (16)$$

The real IEPs may exist at any position along the line connecting the original points ${}^{L_j}\check{\mathbf{p}}_j$ and $\check{\mathbf{p}}_0^{\mathcal{M}}$. As the distance between two points decreases, the midpoint approaches the virtual point with greater confidence. Moreover, the greater the intensity gradient of the IEPs, the more possibility these IEPs are located on real edges with significant texture differences. Therefore, the formula for the weight of IEP w_j^{IEP} is

$$w_j^{\text{IEP}} = \frac{2.0}{1.0 + e^{-\frac{G_j/255}{\Delta l}}} \quad (17)$$

where Δl represents the distance between points ${}^{L_j}\check{\mathbf{p}}_j$ and $\check{\mathbf{p}}_0^{\mathcal{M}}$. To obtain more accurate and reliable weights, the influence of the distance from points to LiDAR is taken into account, and the weighting function is

$$w_j^l = \frac{2.0}{1.0 + e^{-l_j}} \quad (18)$$

where l_j represents the distance from ${}^{L_j}\check{\mathbf{p}}_j$ to the LiDAR frame. w_j^f is the weight of ${}^{L_j}\check{\mathbf{p}}_j$ during feature extraction. Thus, the final weight of ${}^{L_j}\check{\mathbf{p}}_j$ is given by $w_j = w_j^f \times w_j^l$. The designed functions restrict the final weight values within a certain range. A higher weight for feature points indicates higher quality.

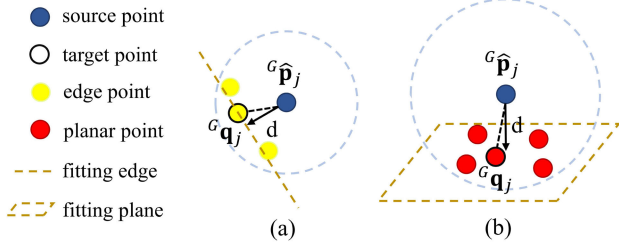


Fig. 5. Overview of point-to-edge and point-to-plane registration. (a) Point-to-edge. (b) Point-to-plane.

3) *Residual Computation*: GEPs and IEPs are treated similar to construct residuals. Assuming the current iteration count of the IEKF is κ , the state at the current iteration update is $\hat{\mathbf{x}}_k^\kappa$. Each feature point ${}^{L_k}\mathbf{p}_j^k$ in $\check{\mathcal{P}}$ can be projected from the LiDAR frame to the global frame

$${}^G\hat{\mathbf{p}}_j^k = {}^G\hat{\mathbf{T}}_{I_k}^\kappa {}^I\hat{\mathbf{T}}_{L_k}^\kappa {}^{L_k}\check{\mathbf{p}}_j^k. \quad (19)$$

For each LiDAR feature point, the plane or edge fit by its nearest neighboring points is assumed to represent the real plane or edge in the environment. Therefore, the residual can be defined as the distance between the feature point ${}^G\hat{\mathbf{p}}_j^k$ in the global frame and the nearest plane or edge in the map, as shown in Fig. 5. Moreover, approximating the measurement (4) by its first-order approximation made at $\hat{\mathbf{x}}_k^\kappa$ leads to

$$\begin{aligned} 0 &= \mathbf{h}_j(\mathbf{x}_k, {}^L\mathbf{n}_j) \simeq \mathbf{h}_j(\hat{\mathbf{x}}_k^\kappa, \mathbf{0}) + \mathbf{H}_j^\kappa \tilde{\mathbf{x}}_k^\kappa + \mathbf{v}_j \\ &= \mathbf{z}_j^\kappa + \mathbf{H}_j^\kappa \tilde{\mathbf{x}}_k^\kappa + \mathbf{v}_j \end{aligned} \quad (20)$$

where $\mathbf{H}_j^\kappa$ is the Jacobian matrix of $\mathbf{h}_j(\hat{\mathbf{x}}_k^\kappa \boxplus \tilde{\mathbf{x}}_k^\kappa, {}^L\mathbf{n}_j)$ with respect to $\tilde{\mathbf{x}}_k^\kappa$, evaluated to zero. $\mathbf{v}_j \in \mathcal{N}(0, \mathbf{R}_j)$ originates from the raw measurement noise ${}^L\mathbf{n}_j$. However, it is necessary to update the LiDAR measurement noise based on the computed weight

$$\hat{\mathbf{R}}_j = \frac{\mathbf{R}_j}{w_j}. \quad (21)$$

4) *State Update*: Combining the prior distribution of $\mathbf{x}$ with the measurement model, the posterior distribution of the state $\mathbf{x}_k$ can be generated, represented equivalently by $\tilde{\mathbf{x}}_k^\kappa$ and its maximum a posteriori (MAP) estimate

$$\min_{\tilde{\mathbf{x}}_k^\kappa} \left(\left\| \mathbf{x}_k \boxminus \hat{\mathbf{x}}_k \right\|_{\hat{\mathbf{P}}_k}^2 + \sum_{j=1}^m \left\| \mathbf{z}_j^\kappa + \mathbf{H}_j^\kappa \tilde{\mathbf{x}}_k^\kappa \right\|_{\hat{\mathbf{R}}_j}^2 \right) \quad (22)$$

where $\|\mathbf{x}\|_{\mathbf{M}}^2 = \mathbf{x}^T \mathbf{M}^{-1} \mathbf{x}$, and m is the total number of valid feature points in the k th frame.

E. Map Management

After obtaining the optimal state $\bar{\mathbf{x}}_k$ for the k th scan, each feature point $\check{\mathbf{p}}_j$ in the feature point cloud scan $\check{\mathcal{P}}$ of the k th frame is projected into the global frame

$${}^G\bar{\mathbf{p}}_j = {}^G\bar{\mathbf{T}}_{I_k} {}^I\bar{\mathbf{T}}_{L_k} {}^{L_k}\check{\mathbf{p}}_j. \quad (23)$$

Based on the ikd-tree, efficient management of map points is achieved, allowing dynamic growth by adding new scans at the

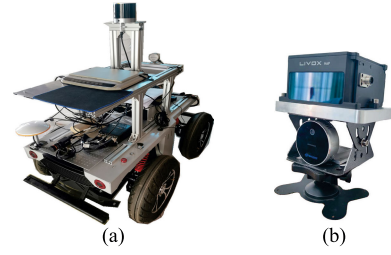


Fig. 6. Sensor configurations. (a) Our unmanned system. (b) Our handheld device.

odometry rate. When the odometry outputs, the feature points of the current scan are transformed to the world frame and then incrementally inserted into the ikd-tree map together with their types. To ensure real-time performance of our method in large-scale environments, only a local region of map points with a length of L around the current LiDAR position in the world frame is maintained on the ikd-tree.

IV. EXPERIMENTAL RESULTS

A. Experimental Setup

In this section, we present comprehensive experiments of the proposed intensity gradient enhanced tightly-coupled LiDAR-inertial odometry (IGE-LIO) method, including validation on both public datasets and our private datasets. We compare IGE-LIO with state-of-the-art and open-source methods, including FAST-LIO, FAST-LIO2, FAST-LIVO [28], and Intensity-SLAM, all of which use their recommended parameters for indoor and outdoor scenarios. To ensure fairness, IGE-LIO maintained consistency with FAST-LIO in geometric feature processing parameters, and the loop-closure module of Intensity-SLAM was removed. The results of all the experiments are the mean values after ten runs. In addition, the numbers of points in $\mathbf{C}_M$ and $\mathbf{C}_N$ were set to 2 in indoor scenarios and 1 in outdoor scenarios to ensure the extraction of an appropriate number of IEPs.

As shown in Fig. 6, there are two sets of sensor suites used for data collection. The first set comprises an unmanned system equipped with an RS-LiDAR-Helios-16P LiDAR, which has a 360° (horizontal) $\times$ 30° (vertical) FoV, at a typical scan rate of 10 Hz. The second set is a handheld device equipped with a Livox HAP LiDAR, which has a 120° (horizontal) $\times$ 25° (vertical) FoV, at a typical scan rate of 10 Hz. Both the setups include a WHEELTEC N100 IMU that outputs data at 100 Hz. For ease of subsequent description, in the Interior_Hall scene, the data collected by the solid-state LiDAR are referred to as Interior_Hall_Solid and the data collected by the rotating LiDAR are referred to as Interior_Hall_Rotation. In addition, we set all the feature points to have equal weights [termed IGE-LIO(w/o fun.)] and reject IEPs [termed IGE-LIO(w/o int.)].

B. Public Dataset

1) *LiDAR Degenerated Experiment*: We conduct comparative experiments on the indoor LiDAR_Degenerated dataset [28] collected by the Livox Avia LiDAR sampled

TABLE I
ABSOLUTE TRANSLATIONAL ERROR (RMSE, METERS) ON THE NTU-VIRAL DATASET

	Eee_01	Eee_02	Eee_03	Nya_01	Nya_02	Nya_03	Sbs_01	Sbs_02	Sbs_03
IGE-LIO(w/o fun.)	0.237	0.198	0.225	0.303	0.211	0.309	0.209	0.222	0.228
IGE-LIO(w/o int.)	0.228	0.197	0.229	0.285	0.198	0.203	0.210	0.221	0.214
IGE-LIO	0.209	0.197	0.217	0.231	0.195	0.194	0.207	0.219	0.212
FAST-LIVO	0.247	0.193	0.227	0.278	0.214	0.228	0.332	0.264	0.227
FAST-LIO2	0.265	0.204	0.232	0.317	0.221	0.248	0.233	0.233	0.236

TABLE II
ABSOLUTE ROTATIONAL ERROR (RMSE, DEGREES) ON THE NTU-VIRAL DATASET

	Eee_01	Eee_02	Eee_03	Nya_01	Nya_02	Nya_03	Sbs_01	Sbs_02	Sbs_03
IGE-LIO(w/o fun.)	4.07	3.37	3.72	3.39	4.14	6.31	3.46	2.95	2.94
IGE-LIO(w/o int.)	5.43	3.68	5.17	4.24	5.21	5.17	5.09	4.27	4.13
IGE-LIO	3.52	2.36	3.18	3.08	3.47	2.75	2.84	2.40	2.59
FAST-LIVO	4.41	2.61	4.22	3.77	4.28	3.59	5.26	3.14	2.71
FAST-LIO2	6.79	6.24	5.76	5.93	5.71	5.53	4.64	6.38	5.75

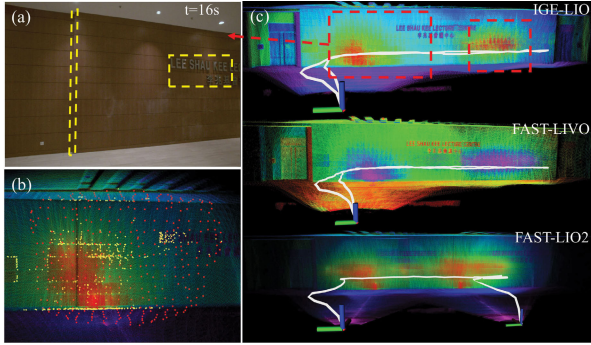


Fig. 7. Performance of IGE-LIO, FAST-LIVO, and FAST-LIO2 in a LiDAR degenerated scene. (a) Area with severe degradation. The yellow dashed box highlights the geometric structures. (b) Extracted feature points, with red points denoting GPPs and yellow points representing IEPs. (c) Results of three algorithms.

at 10 Hz and its built-in IMU. The LiDAR has the non-repetitive scanning pattern with a 70.4° (horizontal) $\times$ 77.2° (vertical) FoV. The scene consists of a wall with a length of about 30 m. If the solid-state LiDAR only observes the side wall, it lacks sufficient constraints in the direction along the wall, making it challenging for the geometric-only odometry to obtain accurate state estimates.

As shown in Fig. 7(c), IGE-LIO successfully returned to the starting position in this degenerated scene. In the area indicated in Fig. 7(a) by the yellow dashed box, only the intersection of planes and protruding letters can provide constraints along the wall direction. However, these two areas are almost in the same plane as the wall, leading to incorrect pose estimates by FAST-LIO2 and failure to return to the starting position. FAST-LIVO, which fuses visual data, can extract features from texture information. IGE-LIO uses LiDAR intensity information to extract additional feature points to handle the degenerated environment, as shown in Fig. 7(b) with the yellow IEPs concentrated on the gaps between wall tiles and protruding letters of different colors from the wall. Compared with FAST-LIO2, our system is more robust and accurate in handling the scene, with the lowest end-to-end drift of 0.04 m. It can achieve a level of accuracy comparable to FAST-LIVO.

Therefore, the intensity gradient information from LiDAR can replace the role played by visual sensors in LiDAR degenerated environments.

2) *Benchmark Datasets*: In this experiment, quantitative evaluations are conducted on all the nine sequences of the NTU-VIRAL dataset [29]. Ground truth is provided by tracking a crystal prism using a total station. The sensors used in the datasets are the horizontal 16-channel OS1 gen1 LiDAR, which has a 360° (horizontal) $\times$ 45° (vertical) FoV at a typical scan rate of 10 Hz, and its internal IMU that outputs angular rate and acceleration measurements at a rate of 100 Hz. Tables I and II present the root-mean-square error (RMSE) results of the translation and rotation parts on the NTU-VIRAL dataset, respectively, and the absolute translational error and absolute rotation error were calculated by evo [30].

In most scenarios, FAST-LIVO achieves higher accuracy compared with FAST-LIO2 due to the fusion of visual information. However, the images can become extremely blurry during rapid drone motion, and low-quality images do not contribute to the accuracy of the odometry. For example, FAST-LIO2 has higher accuracy than FAST-LIVO on the Sbs_01. The proposed method can extract extra additional IEPs without relying on visual information. Therefore, IGE-LIO(w/o fun.) achieves higher accuracy compared with FAST-LIO2 in most scenarios. Furthermore, the introduction of IEPs can impose additional constraints on rotation. For instance, in scenarios such as eee_01, eee_02 and sbs_02, sbs_03, where UAV exhibits fast rotation, FAST-LIO2 incurs larger rotational error, while the proposed method demonstrates significant accuracy improvement. However, the introduction of IEPs also introduces extra uncertainty, which may potentially degrade accuracy. For example, on nya_03, IGE-LIO(w/o fun.) exhibits lower accuracy compared with FAST-LIO2. The weighting functions compensate for the accuracy loss caused by the extra uncertainty, enabling IGE-LIO to surpass both FAST-LIVO and FAST-LIO2 in accuracy. In addition, it is noted that IGE-LIO outperforms FAST-LIO2 in both translation and rotation, and even exceeds FAST-LIVO in most scenarios. These results indicate that the proposed method

TABLE III
END-TO-END TRANSLATION ERRORS ON PRIVATE DATASET (METERS)

	Dataset	Length	IGE-LIO	IGE-LIO(w/o fun.)	IGE-LIO(w/o int.)	FAST-LIO	FAST-LIO2	Intensity-SLAM
Indoor	Interior_Hall_Solid	33.42	0.047	0.059	0.057	0.110	0.070	Fail
	Interior_Hall_Rotation	55.79	0.039	0.042	0.049	0.051	1.202	0.115
	Long_Corridor_Solid	104.79	0.075	1.897	0.957	3.203	0.982	7.292
	Long_Corridor_Rotation	110.986	0.054	0.058	0.059	0.560	0.647	13.773
Outdoor	Campus_1_Solid	306.81	0.066	2.587	0.070	3.410	4.167	15.603
	Campus_1_Rotation	287.963	0.048	0.940	0.915	0.061	2.946	2.082
	Campus_2_Solid	446.326	0.081	0.103	0.534	0.534	0.400	9.822
	Campus_2_Rotation	430.037	0.018	1.585	0.964	0.705	28.868	1.807

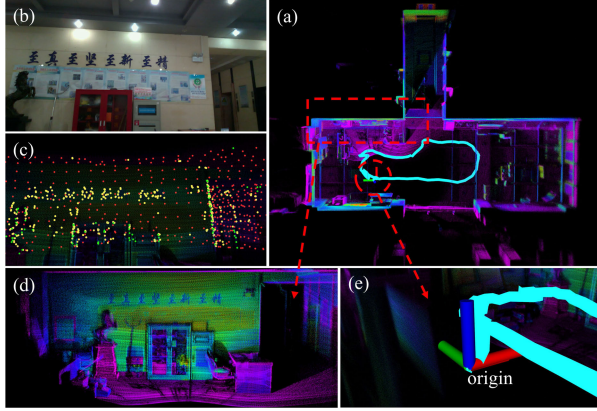


Fig. 8. Mapping results of IGE-LIO using Interior_Hall_Rotation. (a) Constructed point cloud rendered by intensity. (b) and (d) Partially enlarged view of the point cloud and the corresponding real-world snapshot, respectively. (c) Three types of feature points, where red represents GPPs, green represents GEPs, and yellow represents IEPs. (e) Coincidence of the start and end coordinates.

maintains an advantage in typical environments, demonstrating good accuracy and robustness.

C. Private Dataset

1) *Indoor Experiments*: In this experiment, we evaluate our system in two indoor scenes. One scene is an indoor hall with enough geometric structures. Traditional feature extraction methods extract geometric planar features and fewer GEPs from the scene shown in Fig. 8(b). In contrast, our method extracts additional edge points from the boundaries of text signboards and objects with texture differences from the background wall, as shown in Fig. 8(c) as yellow IeEs. This compensates for the insensitivity of geometry-based feature extraction methods to edges in the environment. As shown in Fig. 8(e), the trajectory of IGE-LIO can return to the origin, and the end-to-end error is presented in Table III. This suggests that IGE-LIO has the least global drift. In the results shown in Fig. 9, we observed that on the Interior_Hall_Rotation, FAST-LIO2 extracts insufficient GEPs, resulting in inaccurate pose estimation.

Furthermore, we conducted a robustness experiment in a challenging long corridor environment. The long corridor scene includes a corridor that is 46.5-m long and only 1.9-m wide. To validate the strong robustness of IGE-LIO in this scenario, we translated the solid-state LiDAR in a manner that illuminated the wall almost vertically, as shown in Fig. 10(b).

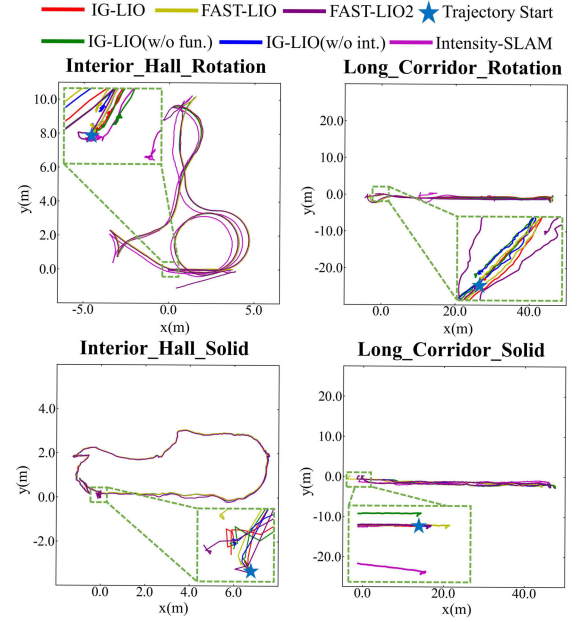


Fig. 9. Results of IGE-LIO on indoor private dataset.

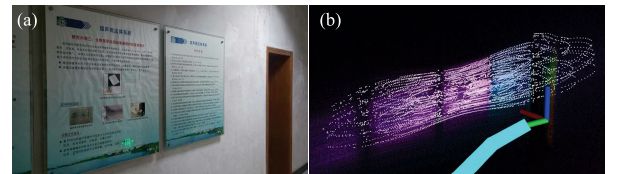


Fig. 10. LiDAR degenerated scenario in the Long_Corridor_Solid dataset. (a) Real-world snapshot of the scene. (b) Corresponding LiDAR scan captured by the solid-state LiDAR.

In this scenario, the billboards are considered to be on the same plane as the wall, leaving only one door frame to provide geometric constraints, as shown in Fig. 10(a). The proposed feature extraction method on IEPs extracts the boundary of the billboards and the white wall, while traditional method can only extract planar features.

In addition, the uncertainty of the IEPs at far distances in narrow corridors increases. To evaluate the uncertainty of IEPs, we used extra IEPs within different distance ranges to calculate poses instead of all IEPs in IGE-LIO (w/o fun.). Moreover, extra IEPs were only used when the LiDAR device was within 16 m from both ends of the long corridor and to ensure a total of 100 IEPs. Furthermore, on the Long_Corridor_Rotation,

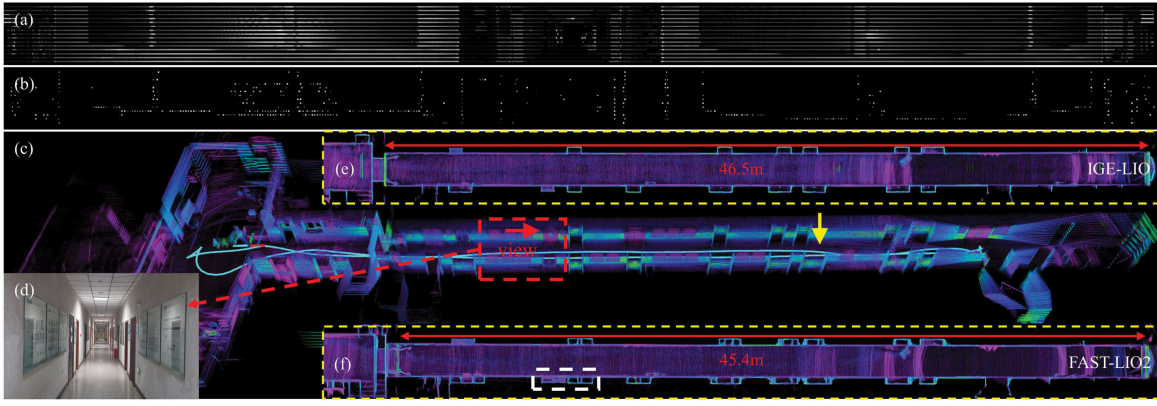


Fig. 11. Mapping results of IGE-LIO using Long_Corridor_Rotation. (a) and (b) Intensity image and intensity gradient image generated by IGE-LIO when the LiDAR is in the scene depicted in (d), which includes a corridor that is 46.5-m long and only 1.9-m wide. For visualization, images (a) and (b) are generated by magnifying the height of the original images five times. (c) Reconstructed point cloud rendered by reflectivity. (d) Real-world snapshot within the red box. (e) and (f) Top view of the local point clouds under IGE-LIO and FAST-LIO2, respectively. (f) Multiple instances of overlap, with the white box marking one of them.

TABLE IV

END-TO-END TRANSLATION ERRORS IN LONG CORRIDOR SCENE USING IEPs WITHIN DIFFERENT DISTANCE RANGES (METERS)

	Long_Corridor_Solid	Long_Corridor_Rotation
IGE-LIO (w/o fun1.)	1.595	0.513
IGE-LIO (w/o fun2.)	1.706	0.624
IGE-LIO (w/o fun3.)	2.452	0.841

IEPs extracted from the point cloud within a semicircular FoV in the forward direction of LiDAR were used to ensure that IEPs originated from the long corridor environment. We divided the distance range into three segments: (10, 20 m), (20, 30 m), and (30, 40 m), denoted as IGE-LIO (w/o fun1.), IGE-LIO (w/o fun2.), and IGE-LIO (w/o fun3.), respectively.

As shown in Table IV, poses calculated using extra IEPs from far distances exhibit lower accuracy, indicating that IEPs at far distances have greater uncertainty, thus implying lower quality. As a result, the accuracy of IGE-LIO (w/o fun.) using Long_Corridor_Solid is lower than that of FAST-LIO2. The weighting functions in IGE-LIO effectively reduce the errors caused by the increased uncertainty. This allows the IEPs to contribute actively to pose estimation, ensuring that IGE-LIO maintains the highest accuracy.

The rotating LiDAR also suffers from severe degradation in the long corridor environment. Although FAST-LIO2 eventually approaches the origin, the overlapping point clouds in Fig. 11(f) suggest that FAST-LIO2 fails to achieve accurate pose estimation, resulting in the error of 1.1 m. Comparing Fig. 11(a) and (b), it can be observed that the IEPs mainly originate from boundaries with significant texture differences in the real world. Fig. 9 shows that IGE-LIO can finally approach or return to the starting position. Furthermore, IGE-LIO demonstrates greater accuracy and robustness when facing the LiDAR degenerated scenario compared with Intensity-SLAM.

2) *Outdoor Experiments:* Here we validate the performance of IGE-LIO in outdoor environments. Fig. 12(b) shows the mapping results on the Campus_1_Rotation dataset. The

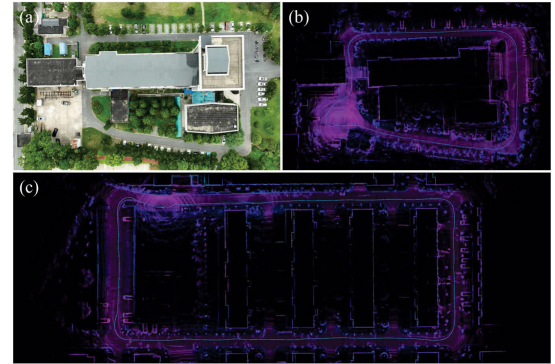


Fig. 12. Mapping results of IGE-LIO on outdoor private dataset. (a) Aerial photograph of Campus_1_Rotation. (b) Point cloud of Campus_1_Rotation. (c) Point cloud of Campus_2_Rotation.

TABLE V

AVERAGE PROCESSING TIME PER SCAN ON PRIVATE DATASET(S)

	IGE-LIO	FAST-LIO2	Intensity-SLAM
Interior_Hall_Solid	0.0324	0.0248	Fail
Interior_Hall_Rotation	0.0402	0.0301	0.0534
Long_Corridor_Solid	0.0274	0.0235	0.0439
Long_Corridor_Rotation	0.0281	0.0194	0.0447
Campus_1_Solid	0.0473	0.0310	0.0750
Campus_1_Rotation	0.0472	0.0325	0.0645
Campus_2_Solid	0.0457	0.0293	0.0728
Campus_2_Rotation	0.0406	0.0327	0.0644

unmanned system returned to the starting position after traveling around 288 m. The drift in this experiment is smaller than 0.017% (0.048 m drift over 288 m trajectory). Moreover, we also validated IGE-LIO in Campus_1_Solid, Campus_2_Rotation, and Campus_2_Solid datasets. The results show that IGE-LIO can achieve better odometry accuracy in outdoor environments, as shown in Table III.

D. Time Analysis

To test the real-time performance of the proposed method on computing platform, we calculated the average processing time per scan of IGE-LIO, FAST-LIO2 and Intensity-SLAM

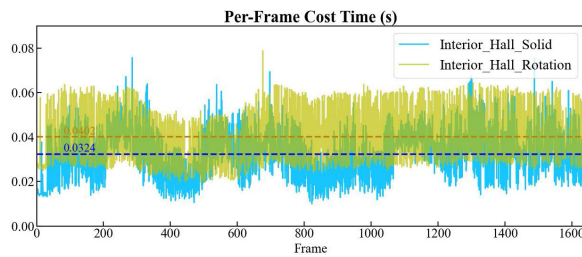


Fig. 13. Per-frame cost time of IGE-LIO.

on Intel 8-core i7-9700 CPU at 3.00 GHz for all private datasets, and the results are shown in Table V. Although Fast-LIO2 achieves optimal time efficiency, the proposed method is fully run in real-time for all the experiments, with competitive performance at around 20 Hz. In addition, the average processing time of IGE-LIO in the Interior_Hall_Rotation and Interior_Hall_Solid datasets is 0.0402 and 0.0324 s, respectively, as shown in Fig. 13. Compared with Intensity-SLAM, which also uses intensity, our method demonstrates significant advantages in terms of time efficiency.

V. CONCLUSION

In this article, we propose IGE-LIO, an intensity gradient enhanced LiDAR-inertial fusion framework, which is more robust and accurate than FAST-LIO2, achieving a comparable level of accuracy with FAST-LIVO that incorporates the camera in degenerated environments. We use the intensity information from LiDAR to extract extra IEPs and fuse the measurements of LiDAR and inertial sensors within an error-state iterated Kalman filter. Extensive quantitative and qualitative experiments show that the introduction of intensity gradient and weighting functions improves the accuracy and robustness of LIO. Furthermore, we demonstrate our system is more stable and accurate in challenging scenarios, including indoor, outdoor, and LiDAR degenerated environments. Future work will introduce back-end optimization and loop closure to improve the global consistency of localization.

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